

Table 1. *G36-HIL* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Medium-sized office building	Chicago, IL, USA	Simulated, integrated with hardware-in-the-loop (HIL)	G36	4	114	5-min	1 day

Table 2. Details of *G36-HIL* fault scenarios. Fault injection times varied in each case, but all faults were simulated using the weather data from August 1st.

Category	Fault abbreviation (CSV file name)	Fault type	Fault intensity	Fault injection duration (start-finish time)	Method of fault imposition
Physical faults	Phy_TSupStp_Max	AHU supply air temperature setpoint	Fixed to maximum value	08:00 – 09:00, 12:00 – 13:00, 16:00 – 17:00	Adjusting the setpoint to its maximum value, consistent with the actual bounds of the AHU controller defined in the building automation system (BAS) server.
	Phy_CooCoiValStuck_0	Cooling coil valve stuck position	Stuck at minimum position (i.e., 0%)	08:00 – 09:00, 12:00 – 13:00, 16:00 – 17:00	Adjusting the valve position through the BAS server.
Cyber-attack faults	Cyber_Device_Reinit_Attack	Device reinitialization attack on the AHU controller	-	08:00 – 08:20, 10:00 – 10:20, 12:00 – 12:20, 14:00 – 14:20, 16:00 – 16:20, 18:00 – 18:20	-
	Cyber_Network_DOS_Attack	Network denial-of-service (DoS) attack on BAS	-	08:00 – 08:20, 10:00 – 10:20, 12:00 – 12:20, 14:00 – 14:20, 16:00 – 16:20, 18:00 – 18:20	-

Additional information:

- Additional helpful document(s):
 - Li, G. et al. Emulation and detection of physical faults and cyber-attacks on building energy systems through real-time hardware-in-the-loop experiments. *Energy and Buildings*, 114596 (2024).
 - Li, G. *et al.* A hardware-in-the-loop (HIL) testbed for cyber-physical energy systems in smart commercial buildings. *Science and Technology for the Built Environment* **30**, 415-432 (2024).
- The value of -123456 in the dataset signifies a missing data point retrieved from the BAS server.
- Please note, since this dataset represents a simulated building, the baseline (fault-free) data was intentionally generated with the same timestamp as the fault data. This was achieved by applying identical outdoor conditions (e.g., outdoor air humidity and temperature), occupancy load, time of day, simulation period, and other related parameters when generating the dataset from the simulated model. This approach allows researchers to compare the corresponding baseline and fault/attack datasets without confounding variations from external factors.
- Fig. 1 shows the schematic of the corresponding AHU system. The system is a single-duct multi-zone VAV AHU connected with five VAV terminal boxes that serve five zones (four exterior and one interior zone) on a single floor.

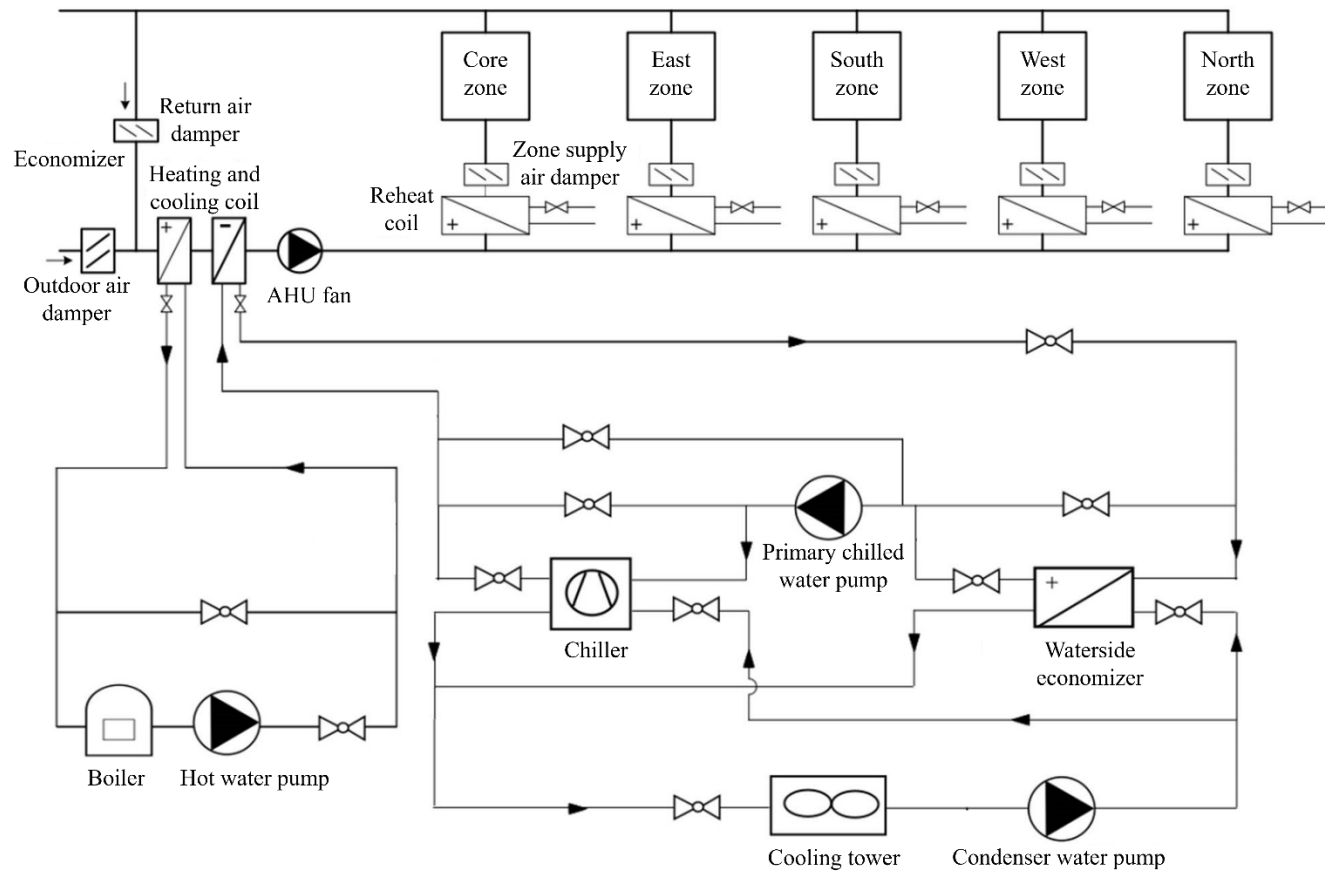


Fig. 1. Schematic of the AHU diagram, used to generate the dataset.